

# Truly Innovative **TECH**

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Tech*

## Tiny lidar with beam steering for 3D sensing

Lumotive has introduced its Meta-Lidar platform, the M30, which is a small and cost-effective 3D sensing solution. It comprises a tiny Lumotive lidar device



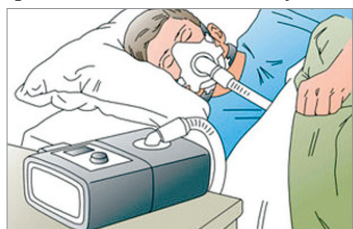
with patented beam steering technology that allows a field of view of  $120^\circ \times 90^\circ$  and  $640 \times 480$  resolution. The M30 also has software-defined scan modes and a reference system design. Powered by Lumotive's highly scalable light control metasurface silicon chips manufactured in a standard CMOS semiconductor process, the Meta-Lidar platform is a scalable solution for a range of size, price, performance, and power requirements - removing barriers to greater proliferation of 3D sensing in automotive, industrial, and consumer applications.

Lumotive

[www.lumotive.com](http://www.lumotive.com)

## Multi-range pressure sensor for sleep apnea devices

Superior Sensor Technology has introduced a new dual low-pressure sensor product series for continuous PAP (CPAP), bi-level PAP (BiPAP), and automatic PAP (APAP) products used for sleep apnea and other breathing sleep disorders. With industry leading



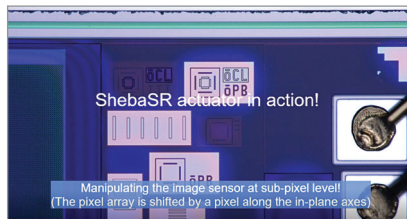
accuracy and performance, the CP series is the industry's first dual, multi-range pressure sensor that integrates a differential pressure sensor and a gauge pressure sensor into one solution. This level of integration in a single module will significantly improve the reliability and performance of PAP devices and offer manufacturers a greatly simplified design and manufacturing process.

Superior Sensor Technology

[www.superiorsensors.com](http://www.superiorsensors.com)

## Innovative pixel-shift imaging technology

Sheba Microsystems has released the ShebaSR, a miniature actuator that can improve the resolution of smartphone camera images by up to nine times. That means nine times less artifacts and nine times more details in the image.



ShebaSR achieves the image enhancement by leveraging pixel-shift super-resolution to obtain a resolution that can be equivalent to up to a 4-time increase in sensor pixel count, as well as reduced noise and artifacts within each pixel using a  $\frac{1}{2}$ -pixel shift. ShebaSR is also remarkably rugged. The product can be subjected to shocks, drops, tumbles, as well as temperature and humidity fluctuations without deterioration in performance.

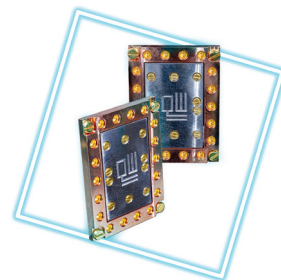
Sheba Microsystems

[www.shebamicrosystems.ca](http://www.shebamicrosystems.ca)

## World's first superconducting quantum processors

Dutch startup QuantWare has launched the world's first commercially available

superconducting processor for quantum computers (QPU) called Soprano. This is the first time superconducting quantum processors have been available off the shelf, a development with the potential



to significantly accelerate the quantum computing revolution. The Soprano is a 5-qubit QPU that has "the fidelities of each qubit to be 99.9 percent, which keeps the error rate manageable." Superconducting qubits are highly customisable, easy to control, and scalable, which practicality makes superconducting QPUs the most likely candidate for quantum computing applications.

Quantware

[www.quantware.eu](http://www.quantware.eu)

## Motion sensor also detects speed and direction

InnoSenT GmbH has developed a new sensor for detecting movements. The IMD-1100 is a small and low-cost standard product, which has a K band radar for typical motion-controlled applications in the field of building automation and security technology. In addition to motion detection, the sensor with CW modulation also enables the speed of

